MSG No. 193.257 $P6'_3/mcm'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-62'm'`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'.m'`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[4,5,6,10,11,12]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[13,14,15,19,20,21]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[16,17,18,22,23,24]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[13,14,15,16,17,18]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[19,20,21,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m'`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,22,23,24]`
2	$[0,\, 0,\, -z]$	`[4,5,6,19,20,21]`
3	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[7,8,9,16,17,18]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 6: Wyckoff site: `6f`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, 0\right]$	`[1,5,19,23]`
2	$\left[0,\, \frac{1}{2},\, 0\right]$	`[2,6,20,24]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, 0\right]$	`[3,4,21,22]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,12,13,18]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,10,14,16]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,11,15,17]`

Table 7: Wyckoff site: `6g`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,8,16,23]`
2	$\left[0, x, \frac{1}{4}\right]$	`[2,9,17,24]`
3	$\left[-x, -x, \frac{1}{4}\right]$	`[3,7,18,22]`
4	$\left[x, x, \frac{3}{4}\right]$	`[4,12,13,21]`
5	$\left[-x, 0, \frac{3}{4}\right]$	`[5,10,14,19]`
6	$\left[0, -x, \frac{3}{4}\right]$	`[6,11,15,20]`

Table 8: Wyckoff site: `8h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3]`
2	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[4,5,6]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	`[10,11,12]`
5	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[13,14,15]`
6	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	`[16,17,18]`
7	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[19,20,21]`
8	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[22,23,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `..2`

No.	position	mapping
1	$[x, 2x, 0]$	`[1,5]`
2	$[-2x, -x, 0]$	`[2,6]`
3	$[x, -x, 0]$	`[3,4]`
4	$\left[x, -x, \frac{1}{2}\right]$	`[7,12]`
5	$\left[x, 2x, \frac{1}{2}\right]$	`[8,10]`
6	$\left[-2x, -x, \frac{1}{2}\right]$	`[9,11]`
7	$\left[-x, x, \frac{1}{2}\right]$	`[13,18]`
8	$\left[-x, -2x, \frac{1}{2}\right]$	`[14,16]`
9	$\left[2x, x, \frac{1}{2}\right]$	`[15,17]`
10	$[-x, -2x, 0]$	`[19,23]`
11	$[2x, x, 0]$	`[20,24]`
12	$[-x, x, 0]$	`[21,22]`

Table 10: Wyckoff site: `12j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	`[1,8]`
2	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	`[3,7]`
4	$\left[x,\, x-y,\, \frac{3}{4}\right]$	`[4,12]`
5	$\left[-x+y,\, y,\, \frac{3}{4}\right]$	`[5,10]`
6	$\left[-y,\, -x,\, \frac{3}{4}\right]$	`[6,11]`
7	$\left[x-y,\, x,\, \frac{3}{4}\right]$	`[13,21]`
8	$\left[-x,\, -y,\, \frac{3}{4}\right]$	`[14,19]`
9	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	`[15,20]`
10	$\left[x-y,\, -y,\, \frac{1}{4}\right]$	`[16,23]`
11	$\left[y,\, x,\, \frac{1}{4}\right]$	`[17,24]`
12	$\left[-x,\, -x+y,\, \frac{1}{4}\right]$	`[18,22]`

Table 11: Wyckoff site: `12k`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,23]`
2	$[0,\, x,\, z]$	`[2,24]`
3	$[-x,\, -x,\, z]$	`[3,22]`
4	$[x,\, x,\, -z]$	`[4,21]`
5	$[-x,\, 0,\, -z]$	`[5,19]`
6	$[0,\, -x,\, -z]$	`[6,20]`
7	$\left[-x,\, -x,\, \frac{1}{2}-z\right]$	`[7,18]`
8	$\left[x,\, 0,\, \frac{1}{2}-z\right]$	`[8,16]`
9	$\left[0,\, x,\, \frac{1}{2}-z\right]$	`[9,17]`
10	$\left[-x,\, 0,\, z+\frac{1}{2}\right]$	`[10,14]`
11	$\left[0,\, -x,\, z+\frac{1}{2}\right]$	`[11,15]`
12	$\left[x,\, x,\, z+\frac{1}{2}\right]$	`[12,13]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$[x,\, x-y,\, -z]$	`[4]`
5	$[-x+y,\, y,\, -z]$	`[5]`
6	$[-y,\, -x,\, -z]$	`[6]`
7	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	`[7]`
8	$\left[x,\, y,\, \frac{1}{2}-z\right]$	`[8]`
9	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	`[9]`
10	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[16]
17	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[17]
18	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[18]
19	$[-x,\, -y,\, -z]$	[19]
20	$[y,\, -x+y,\, -z]$	[20]
21	$[x-y,\, x,\, -z]$	[21]
22	$[-x,\, -x+y,\, z]$	[22]
23	$[x-y,\, -y,\, z]$	[23]
24	$[y,\, x,\, z]$	[24]